UGP MID-TERM EVALUATION PRESENTATION 2026-27-I

PHYSICS INFORMED VLM-BASED ROBOT NAVIGATION

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Github

Task-1: Finding the Right Simulator for Quadruped Navigation

TASK 1 EVALUATION MATRIX

Simulation Platform Trade-Off & Selection Analysis

Hardware Constraint: Laptop RTX 3050 (4 GB VRAM) • Target: Unitree Go2 Locomotion in Photorealistic Scenes

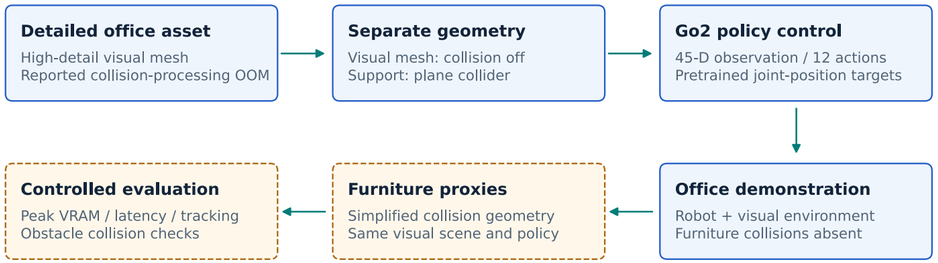
BUDGET ENFORCEMENT
4 GB VRAM Ceiling
Strict zero-OOM constraint

Evaluation Criterion 5 Architectural Axes	★ SELECTED : VISUAL **Genesis** Decoupled Physics & RayTracer	Isaac Sim Omniverse / PhysX 5	Unreal / Unity Game Engine Pipelines	AI Habitat Habitat-Sim / Matterport	★ SELECTED : PHYSICS **MuJoCo** DeepMind Contact Engine
Quadruped Physics Contact stability & Go2 model	✓ **Excellent** Fast rigid-body, Go2 URDF	✓ **Excellent** PhysX 5 GPU pipeline	! **Hard to Bridge** High IPC / sync latency	✗ **Weak Contact** Not tuned for legged robots	✓ **Best-in-Class** Convex contact dynamics
Visual Realism PBR lighting & textures	✓ **Path Tracing** Native raytracer backend	✓ **Very High** Full RTX raytracing	✓ **Very High** Lumen / Nanite assets	✓ **High** Scanned Matterport scenes	✗ **Low** Basic OpenGL rasterizer
RTX 3050 Fit (4 GB) Memory ceiling compliance	✓ **Runs Smoothly** Lightweight preview mode	✗ **OOM Crash** Hard min: ≥ 16 GB VRAM	✗ **>100 GB Disk** Heavy VRAM / storage tax	✓ **Yes** Moderate RAM/VRAM load	✓ **Highly Efficient** < 1 GB total footprint
RL Python API PyTorch & policy loop ease	✓ **Native & Direct** Zero IPC; PyTorch tensors	! **Heavy Stack** Omniverse runtime layers	✗ **Complex Bridge** Requires custom sockets	! **Medium** Custom Gym wrappers	✓ **Native & Direct** Gymnasium / MJX support
Setup Effort Installation & dependencies	✓ **pip install** Single command setup	✗ **Very High** NVIDIA Omniverse app	✗ **Very High** Gigabyte SDK downloads	! **Medium** Conda & C++ builds	✓ **pip install** Fast Python wheel

ARCHITECTURAL DECISION — **Complementary Dual-Simulator Architecture**

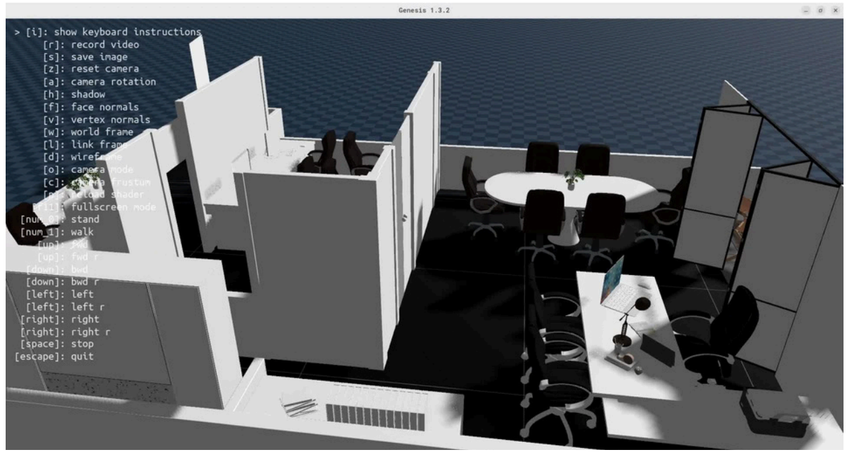
- **Genesis — Chosen for Photorealistic Environment Simulation**
  Loads 3D office (.glb) with decoupled physics plane; fits strictly within 4 GB VRAM.
- **MuJoCo — Chosen for Locomotion Contact Dynamics & Sim-to-Real**
  Gold-standard foot contacts for SvanM2 quadruped MoE policy training (Task 3).

Current prototype and the collision-proxy experiment that remains pending



Dashed boxes: pending. Reported OOM has no saved numerical memory profile.

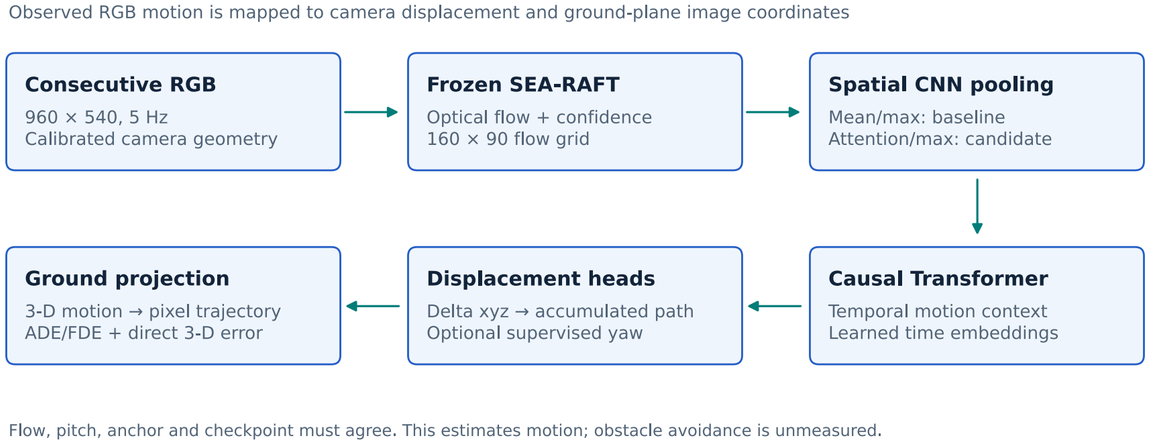
- **Objective:** Identify the best simulation stack for quadruped navigation combining photorealistic indoor environments with high-fidelity contact physics, constrained to a 4 GB VRAM budget (RTX 3050).
  - **Decision**: two simulators, each for its strength:
    - **Genesis**-photorealistic indoor scenes, Go2 integration, low-VRAM rendering, and environment loading.
    - **MuJoCo-**Good contact physics for locomotion training, actuator modelling, and sim-to-real transfer (used in Task 3).

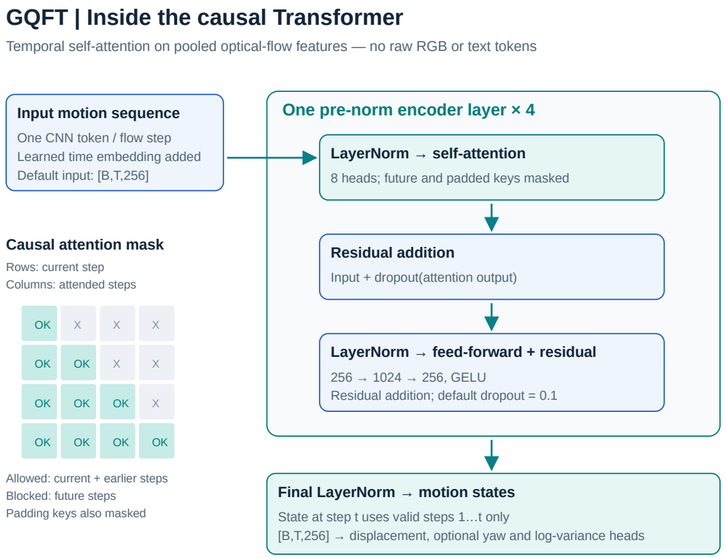


Task_1_Simulator_Evaluation/run_go2.py

Full evaluation with hardware specs

Task-2: Improving GQFT Algorithm





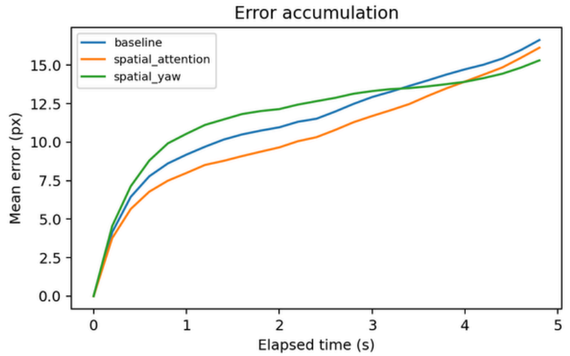
Encoder: 4 layers, 8 heads, 256-D features; past/current attention only. The query enters projection/validity after motion encoding.
SEA-RAFT — ECCV 2024.

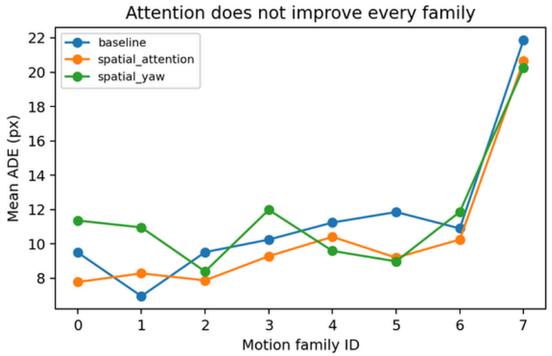
- Estimate camera motion and ground-plane pixel trajectories from observed optical flow.
- **Objective**: Reproduce GQFT and reduce trajectory error with a consistent evaluation protocol.
- **Attempts:** preserve the released **mean+max pooling baseline**; replace the mean term with **learned regional attention** while retaining max pooling; add an **auxiliary yaw head** to supervise heading-related features. Historical flip experiments are retained as discontinued attempts and are excluded from the planned controlled comparison.
- **Analysis → change**: regional-flow probes show stronger yaw cues (0.1574 → 0.1223 rad MAE), supporting spatial aggregation; yaw supervision tests heading-sensitive features.
- **Results:** all three variants have recorded comparisons, but their gains depend on the split and motion family. Attention improves matched validation ADE; Attention + Yaw has the best saved fixed-history test result. Neither establishes a universal improvement across protocols.

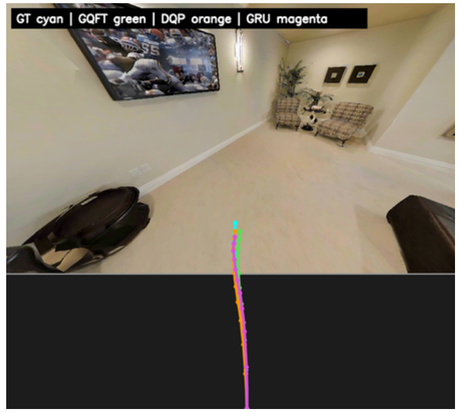
Task-2: Improving GQFT Algorithm (Comparison results — gains and failure cases)

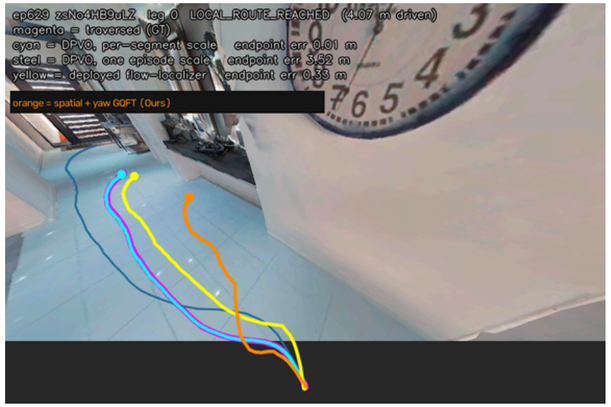
Variant	Val ADE	Test ADE	Test FDE
Baseline	11.263	9.631	15.847
Attention	**10.283**	10.091	16.591
Attention + Yaw	11.636	**8.783**	**13.323**

Matched validation: 104 windows, one query, history 24, CPU float32





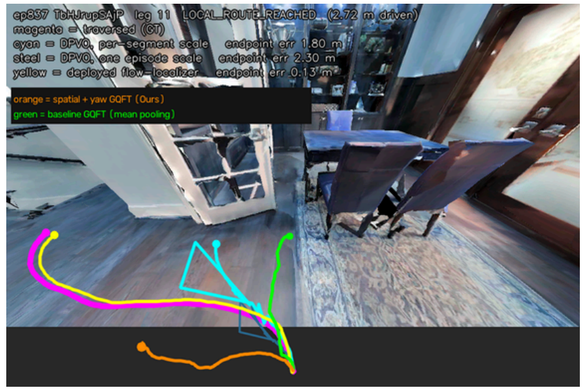


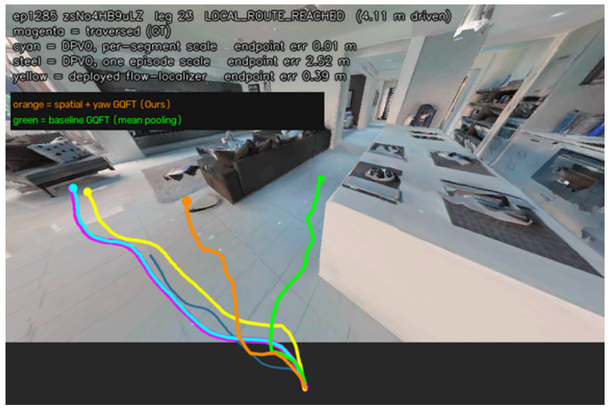


- **Objective:** Determine whether a model improves the whole trajectory, only its endpoint, or particular motion types. Use ADE for average path-point error and FDE for endpoint error; report both rather than choosing the more favorable metric.
- **Approach:** Matched CPU validation plus separately reported saved fixed-history test results.
- **Results:** Attention improves validation ADE 8.7%; Attention + Yaw improves saved test ADE/FDE 8.8% / 15.9%.
- **Analysis → decision:** Attention worsens **constant-arc ADE from 6.961 to 8.285** px despite its overall validation gain, and it fails the predefined history-selection gate. Yaw improves validation FDE while worsening validation ADE. These patterns motivate motion-family analysis and a fixed evaluation protocol before selecting an architecture.

Task-2: Improving GQFT Algorithm (Episode cases that still need improvement)







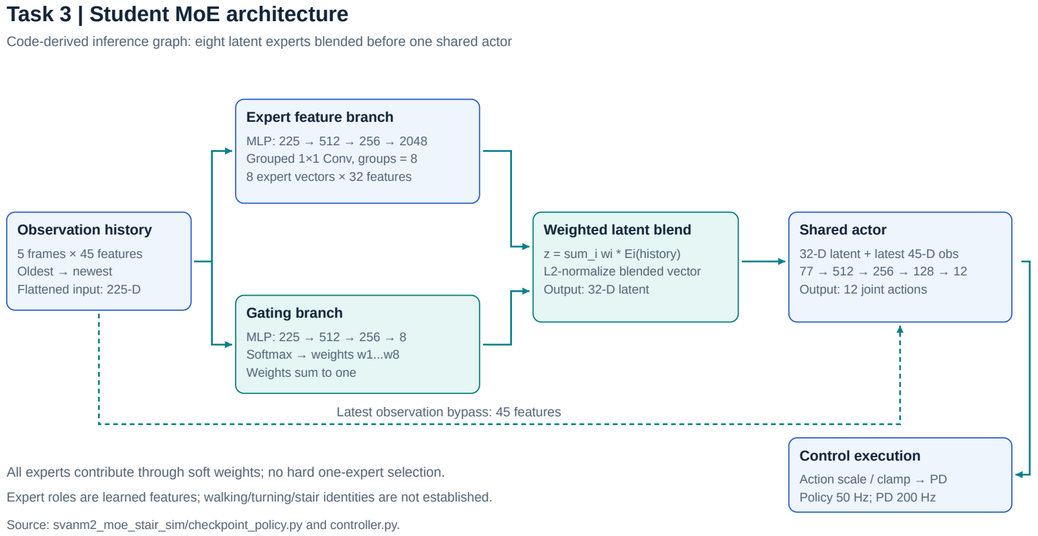
Historical episode overlays—adapter/checkpoint provenance unresolved.

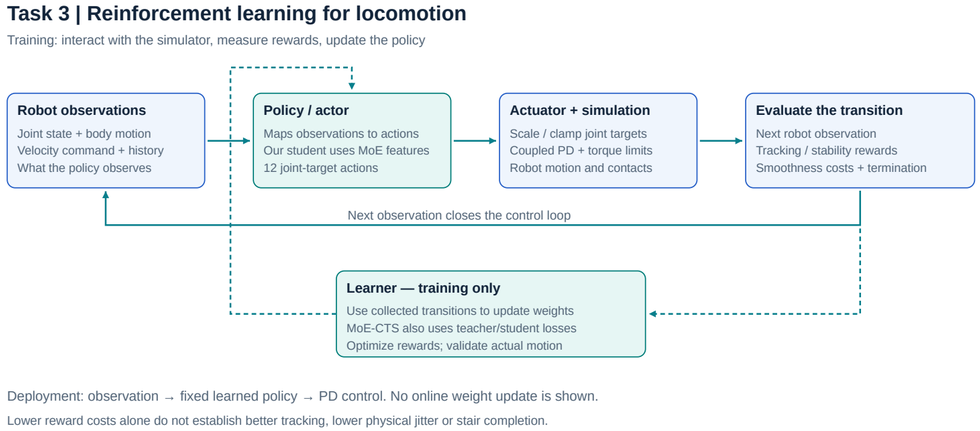
- **Residual path-length, curvature and heading disagreement in saved episode overlays.**
- **Objective:** identify behavior that aggregate ADE/FDE can hide, and define what corrected episode evaluation must measure.
- **Approach:** Inspected a near-straight segment, a curved segment, and a stronger turning case against the traversed-path reference.

Episode case	Visible issue	Improvement to investigate
ep837 · leg 7 — near-straight	Orange follows the route direction but ends short of the traversed path	Path-length/endpoint accuracy; consistent motion scale and segment joining
ep1285 · leg 23 — curved	Orange and green differ from the reference in curvature and endpoint location	Heading and curvature consistency during turns
ep837 · leg 11 — turning	Orange enters the extended canvas and visibly departs from the traversed route	Turning-case evaluation; verify projection/anchor and accumulated motion before attributing failure to the model

- **Analysis → next step:** regenerate these cases through the corrected flow/camera/checkpoint adapter. Measure endpoint drift, path-length error, and turning error with one common protocol; record checkpoint hashes and camera/anchor conventions.

Task-3: Locomotion Policy (RL-MoE)—Fixing the Locomotion Jitter in MuJoCo



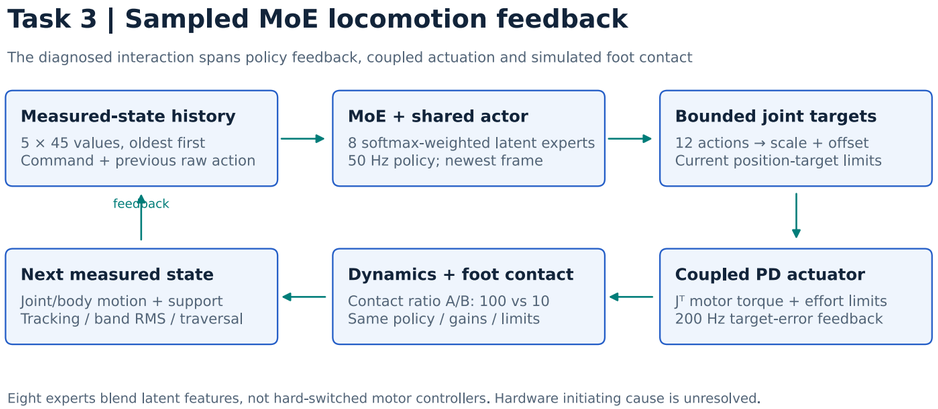


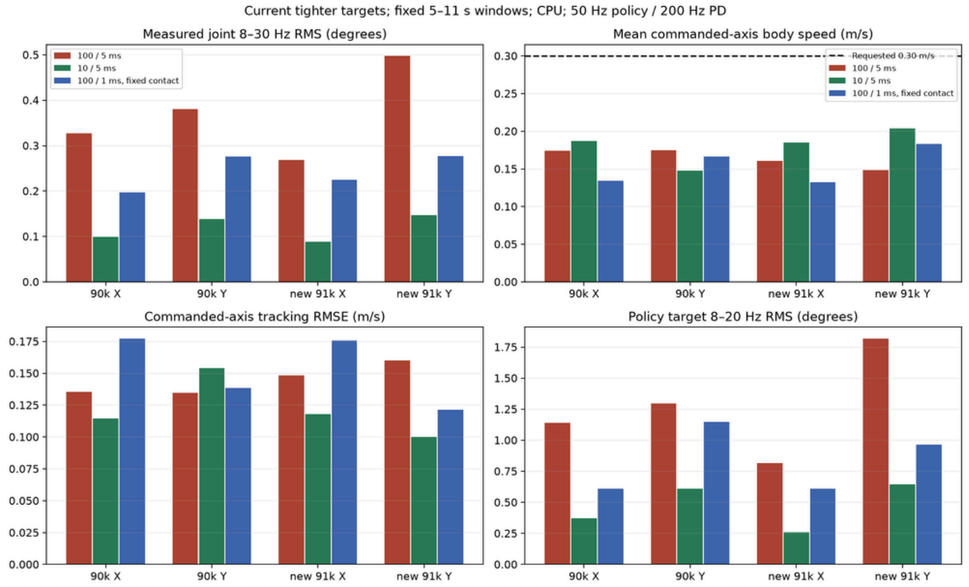
RL-MoE :

- **Reinforcement learning (RL):** learn a policy through interaction with a simulator. Robot observations produce actions; motion produces rewards; the learner updates the policy to improve long-term performance.
- **Our objective:** track velocity commands while maintaining balance, foot support, and smooth joint motion.
- **Mixture of Experts (MoE):** several feature experts process the same observation history. A gating network assigns soft weights and blends their outputs; one shared actor then predicts joint actions.
- **Our architecture:** 5 × 45-D history → 8 experts × 32-D features → weighted 32-D latent + latest 45-D observation → 12 joint actions.
- **Control:** scale/clamp actions into position targets; execute coupled PD at 200 Hz, with policy updates at 50 Hz. At deployment, the learned weights stay fixed.

CTS — IEEE RA-L 2024 · Eight latent experts, one shared actor.

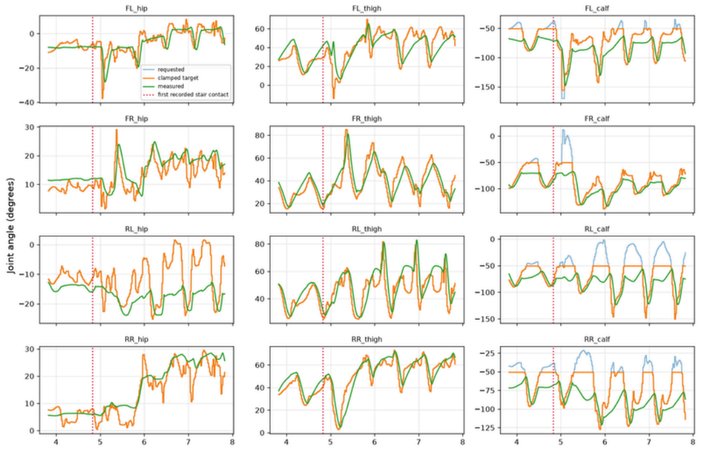
Task-3: Locomotion Policy (RL-MoE)—Locomotion jitter- attempts, contact fix

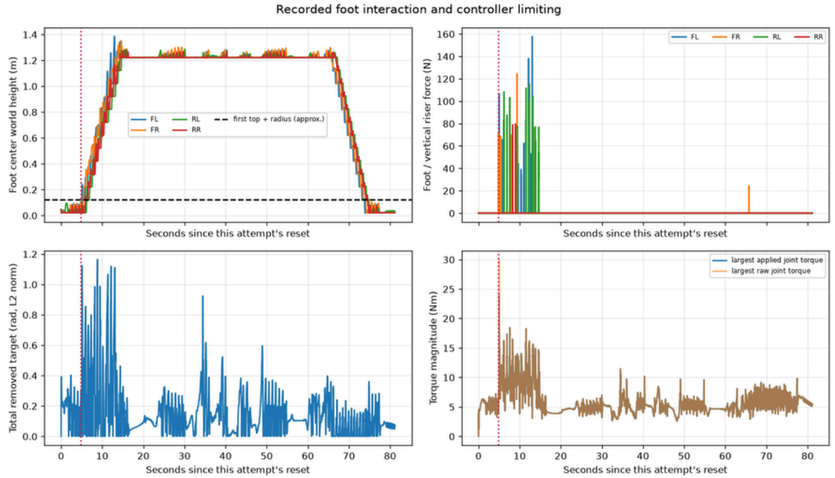




- **What is it about? / Objective:** reduce joint vibration while preserving commanded speed, heading, and stair support in SvanM2.
- **Approach: 225-D history → eight blended latent experts → 12 PD targets**; policy **50 Hz**, PD/physics **200 Hz**. Analyze target/angle traces, reward continuations, and matched simulation interventions.
- **Attempts and outcome: action rate,** target smoothness/saturation costs, and gain/sensing/delay probes. Bundle90500 lowered training second-difference/saturation costs **6.57%/59.65%**, but completed 0/12 staircases.
- **Analysis → fix:** reward gains did not reliably improve motion, and train/viewer contact conventions differed → expose a runtime contact override and test impratio 100 → 10, holding policy, gains, and limits fixed per pair.
- **Results:** four primary flat comparisons reduce joint-band RMS 63.5–70.5%; current stair cases improve 0/6 → 3/6. Original-policy sideways speed falls 0.175 → 0.148 m/s, revealing a stability/tracking tradeoff.
- **Next:** verify deployed bounds/scales/gains/history/delay; measure actuator lag and contact slip. Prioritize calibration or isolated reward ablations, reporting vibration, velocity error, heading, clipping and foot support together.

Task-3: Locomotion Policy (RL-MoE)—Audit Fixes, Improvements, and Analysis





- **Fixes from the audit**
  - **Target limits:** calf upper +0.50000 → −0.88776 rad; hip ±1.0472 → ±0.735 rad. These corrections are present in the current controller.
  - **Policy/control conventions:** verify checkpoint and history through replay; preserve raw-action history and coupled PD instead of introducing unsupported changes.
- **Measured improvements**
  - **Replay verification**: reproduced **30,175 recorded policy outputs**; wrong weights/history were not supported explanations.
  - **Controlled contact intervention:** under corrected tight bounds, the ratio 100 → 10 reduced flat joint-band **RMS 63.5–70.5%.**
  - **Remaining tradeoff:** original-policy sideways speed fell 0.175 → 0.148 m/s. Bounds correction alone did not solve jitter or stair failure; hardware improvement remains unverified.
- **Analysis method**
  - Record each interactive attempt with commands, requested/clamped targets, measured joints, torques, and foot contacts.
  - Use the dashboard to align signals to stair contact and compare approach, impact/recovery, ascent, and settling.

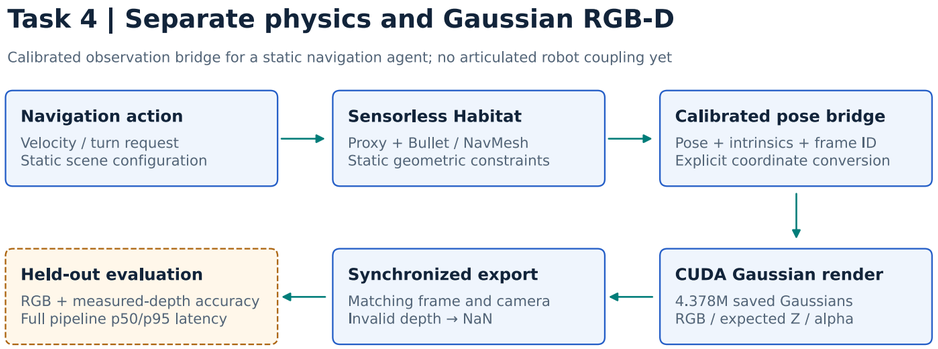
Task-4:Fixing the 3D Models in the RxR dataset



- **Objective:** Improve distorted indoor objects and inspect the repaired scene from calibrated views and first-person room views.
- **Approach:** photo-guided mesh fitting, licensed substitutes, PBR material preservation, and synchronized 360° before/after panoramas at five camera positions.
- **Analysis → fixes:** shape/support errors → calibrated fitting; misclassified mesh children → hierarchy-aware materials; missing handset/dock → restore its active mesh and correct object-name selection.
- **Results:** The latest v6 records verify 23 calibrated comparisons and 30 room/model selections. The original toaster keeps 11 parts and its original PBR materials.



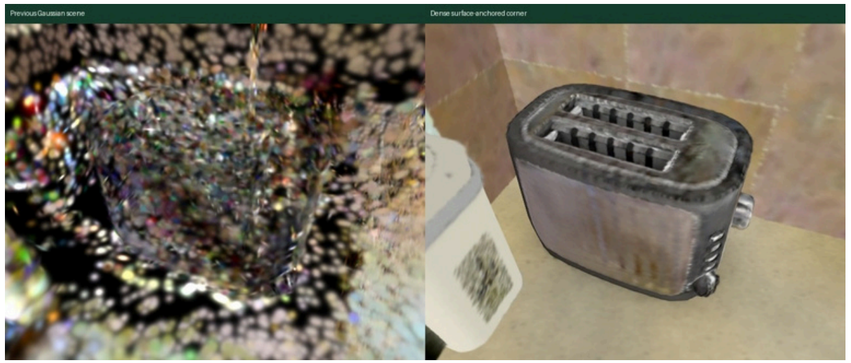
Task-4: Gaussian RGB-D—coverage versus reconstruction quality



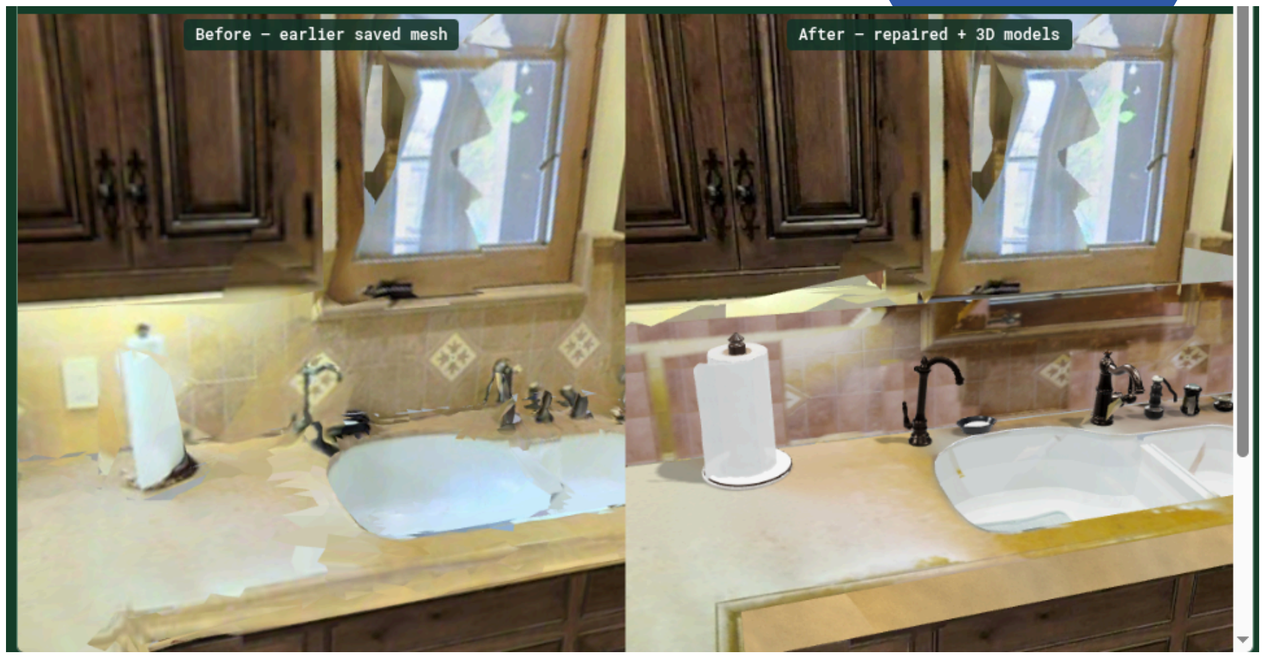
99.0068% valid depth is one-frame coverage. 12.0576 ms is raster-kernel time, not end-to-end RGB-D latency.



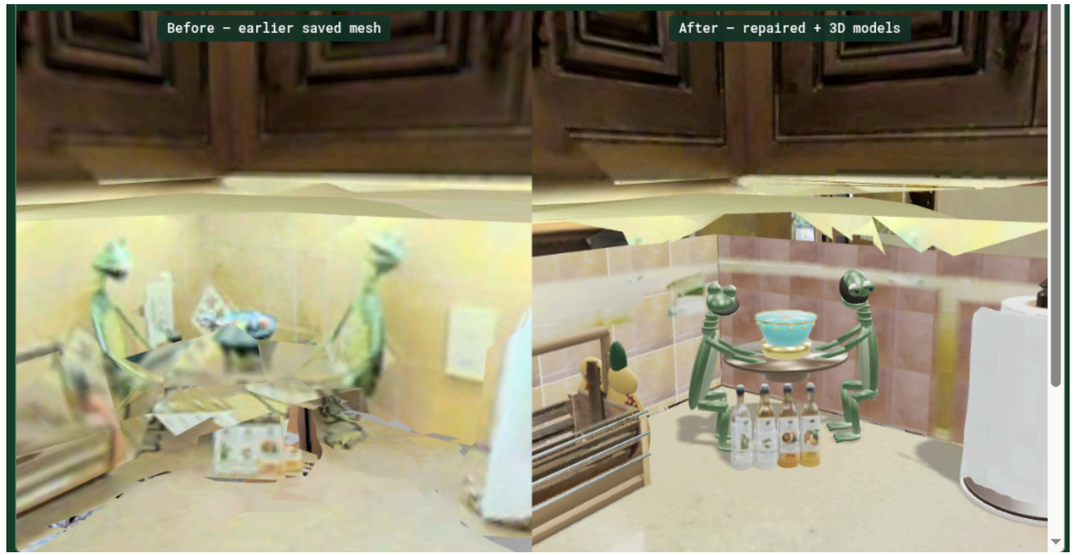
- **What is it about?** Pose-calibrated RGB-D rendering of the repaired kitchen.
- **Objective:** provide useful visual observations alongside a lightweight collision/NavMesh scene.
- **Approach:** separate static physics geometry from the 3D Gaussian renderer.
- **Analysis → change:** floating/irregular corner geometry motivated dense fixed-surface sampling and appearance refinement. Weak held out views showed that matching reference cameras was insufficient. Wider missing coverage then motivated an all-kitchen expansion; its final held-out accuracy remains unmeasured.
- **Results:** 4.378 M Gaussians, 99.01% valid depth in one saved frame, 12.06 ms raster kernel. Earlier held-out PSNR remains weak: 7.28 / 10.48 dB.



Task-4: RxR Dataset fixing(Sample Pictures)







Thank You